



INDEFINITE INTEGRATION

[INTEGRATION BY PARTS]



01. The value of $\int_{-\pi/3}^{\pi/3} \frac{x \sin x}{\cos^2 x} dx$ is [NIMCET 2015, 2023]
 (a) $\frac{1}{3}(4\pi+1)$ (b) $\frac{4\pi}{3} - 2\log \tan \frac{5\pi}{12}$ (c) $\frac{4\pi}{3} + \log \tan \frac{5\pi}{12}$ (d) $\frac{4\pi}{3} - \log \tan \frac{5\pi}{12}$
02. If $\int x \sin x \sec^3 x dx = \frac{1}{2} \left[f(x) \sec^2 x + g(x) \left(\frac{\tan x}{x} \right) \right] + c$ then which of the following is true [NIMCET 2023]
 (a) $f(x) - g(x) = 0$ (b) $f(x) \cdot g(x) = 0$ (c) $f(x) + g(x) = 0$ (d) $f(x) + g(x) = 1$
03. $\int f(x) dx = g(x)$ then $\int x^5 f(x^3) dx$ [NIMCET 2023]
 (a) $\frac{1}{3} x^3 g(x^3) - 3 \int x^4 g(x^3) dx + c$ (b) $\frac{1}{3} x^3 g(x^3) - \int x^2 g(x^3) dx + c$
 (c) $\frac{1}{3} x^3 g(x^3) - \int x^3 g(x^3) dx + c$ (d) None of these
04. $\int e^x (\sinh x + \cosh x) dx = ?$ [NIMCET 2021]
 (a) $e^x \operatorname{sech} x + C$ (b) $e^x \cosh x + C$ (c) $\sinh 2x + C$ (d) $\cosh 2x + C$
05. Evaluate $\int e^x \left(\frac{1 + \sin x \cos x}{\cos^2 x} \right) dx$ [NIMCET 2020]
 (a) $e^x \cos x + c$ (b) $e^x \sec x \tan x + c$ (c) $e^x \tan x + c$ (d) $e^x \cos 2x - 1 + c$
06. $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to [NIMCET 2016]
 (a) $\frac{x e^x}{1 + x^2} + C$ (b) $\frac{x}{(\log x)^2 + 1} + C$ (c) $\frac{\log x}{(\log x)^2 + 1} + C$ (d) $\frac{x}{x^2 + 1} + C$
07. If $\int e^x (f(x) - f'(x)) dx = \phi(x)$, then the value of $\int e^x f(x) dx$ is [NIMCET 2015]
 (a) $\phi(x) + e^x f(x)$ (b) $\phi(x) - e^x f(x)$ (c) $1/2 [\phi(x) + e^x f(x)]$ (d) $1/2 [\phi(x) + e^x f'(x)]$
08. If $\int \frac{x e^x}{\sqrt{1+e^x}} dx = f(x) \sqrt{1+e^x} - 2 \log \frac{\sqrt{1+e^x} - 1}{\sqrt{1+e^x} + 1} + C$, then $f(x)$ is [NIMCET 2014]
 (a) $2x - 1$ (b) $2x - 4$ (c) $x + 4$ (d) $x - 4$
09. The value of $\int \sqrt{x} e^{\sqrt{x}} dx$ is equal to [NIMCET 2014]
 (a) $2\sqrt{x} - e^{\sqrt{x}} - 4\sqrt{x} e^{\sqrt{x}} + C$ (b) $(2x - 4\sqrt{x} + 4)e^{\sqrt{x}} + C$ (c) $(2x + 4\sqrt{x} + 4)e^{\sqrt{x}} + C$ (d) $(1 - 4\sqrt{x})e^{\sqrt{x}} + C$

Answer Key

01. (b) 02. (c) 03. (b) 04. (b) 05. (c) 06. (b) 07. (c) 08. (b) 09. (b)